Interfacing Macaque Cortex with Language Models for Generative Neural Readout

Victoria Bosch*[1], Daniel Anthes*[1], Adrien Doerig[2], Peter König[1], & Tim C Kietzmann[1]

1 - Osnabrück University
2 - Freie Universität Berlin.

*equal contribution
Correspondence: vbosch@uos.de

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Summary

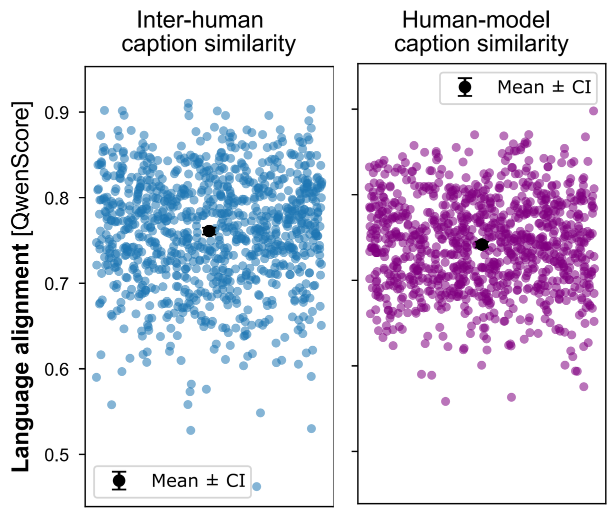
Recent advances in neural decoding have moved beyond categorical readouts toward richer, semantically grounded representations, for example by mapping brain activity elicited by natural scenes to language embeddings of image captions. While effective, such approaches remain limited to static representations. To enable more flexible and interactive decoding of neural data, our recent work has proposed **interfacing brain activity directly with large language models** (***CorText***, Bosch et al., 2025), **allowing neural signals to condition open-ended language generation**. CorText has so far been restricted to time-averaged human fMRI signals.

Here, we extend the approach to **intracranial recordings from macaque visual cortex** during passive viewing in the THINGS dataset (Papale et al., 2025). **We show that time-resolved neural activity is decodable into natural language, enabling generative readout of visual content.** These preliminary results demonstrate that LLM-based neural decoding interfaces generalize beyond human fMRI and suggest the presence of structure in non-human primate visual representations that can be aligned with natural language embeddings.

Datasets

THINGS ventral stream spiking dataset (TVSD)
TVSD consists of electrophysiology recordings from two macaques collected during passive viewing 22k natural images from the THINGS dataset. Multi unit activity (MUA) was simultaneously recorded from 16 64 channel Utah arrays implanted across V1, V4, and IT. For presentation to the model, MUA is averaged in 10ms bins, and z-scored on the training split separately for each session, channel, and included time point. We use the train-test split provided in the original paper, where 100 test stimuli were repeated 30 times.

Visual Question-Answering Dataset
We generate 5 captions and 5 question-answer pairs for each stimulus using the Kimi 2.6 vision-language model. When generating question-answer pairs, the LLM was additionally instructed to categorize each question into one of a set of provided categories (numerosity, colour, position, actions, yes/no, other). The generated captions are combined with a set of predefined caption questions, such as "*Give a concise and descriptive caption of this image.*". To validate the dataset quality, we generate captions for 1000 COCO images, for each of which 5 human captions are available. We calculate the inter-human caption similarity and the human-model caption similarity using QwenScore.

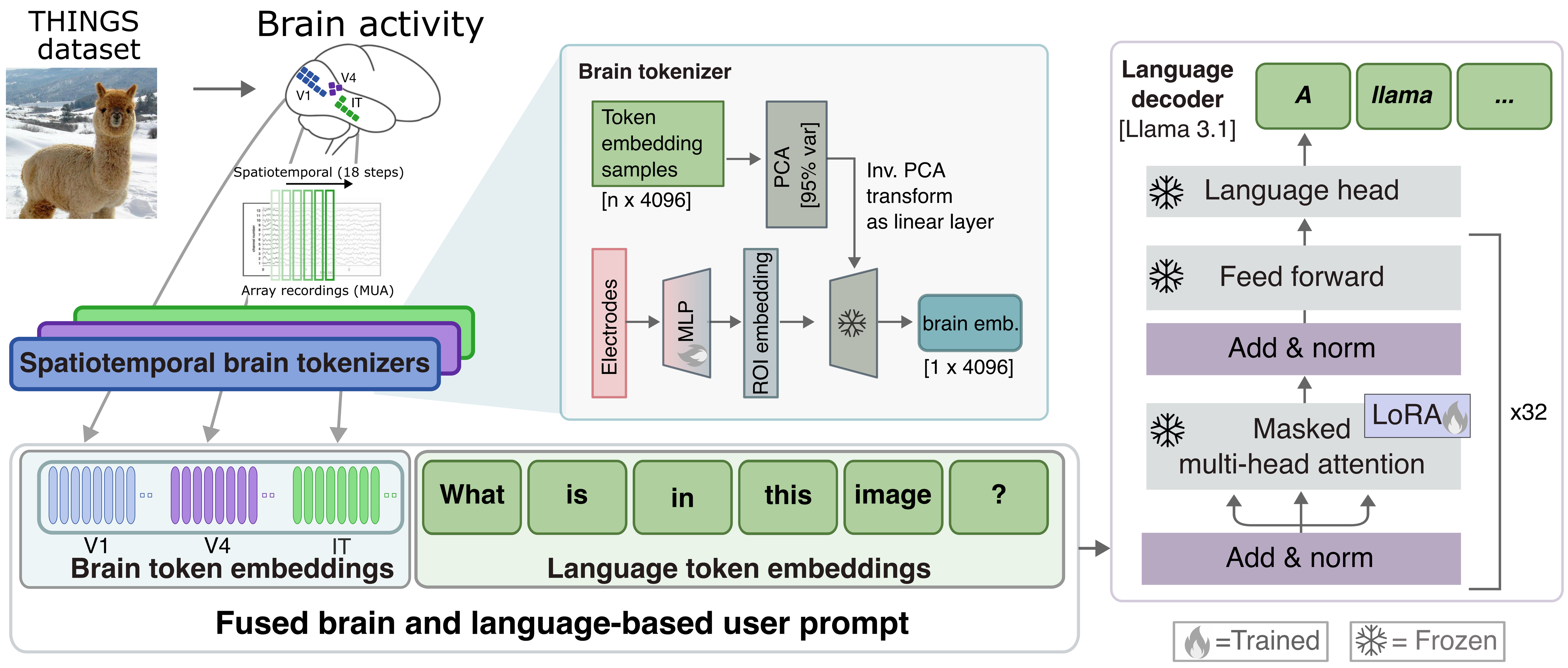


CorText Architecture

CorText is a brain-language model based on a pretrained language transformer (Llama-3.1-instruct 8B; Dubey et al., 2024), and trainable brain tokenizers.

Spatiotemporal brain tokenizers
Recordings are parcellated both in space and time (Anthes et al., 2026): each parcel contains data from one Utah array, and one timepoint between 20-190ms, in 10ms steps (16 arrays x 18 time points = 288 parcels total). Each parcel is embedded using a 2 layer MLP (hidden size of 32; GeLU).

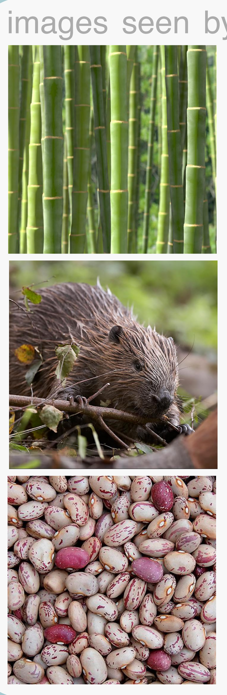
Training (end-to-end):
Supervised finetuning using CE between generated and true answers.
Pretraining (20 ep.): brain tokenizers only, freezing LLM. Learni mapping from brain data into language embeddings.
Finetuning (5 ep.): training the brain tokenizers and LLM using LoRA.



Neural decoding results

General scene descriptions

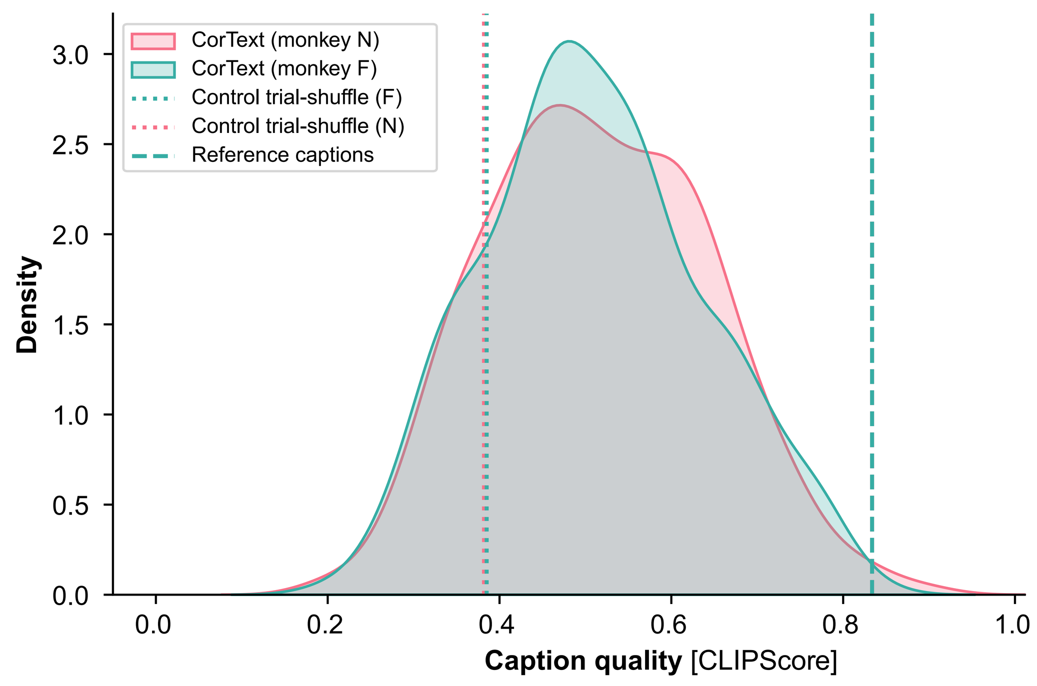
images seen by monkey F

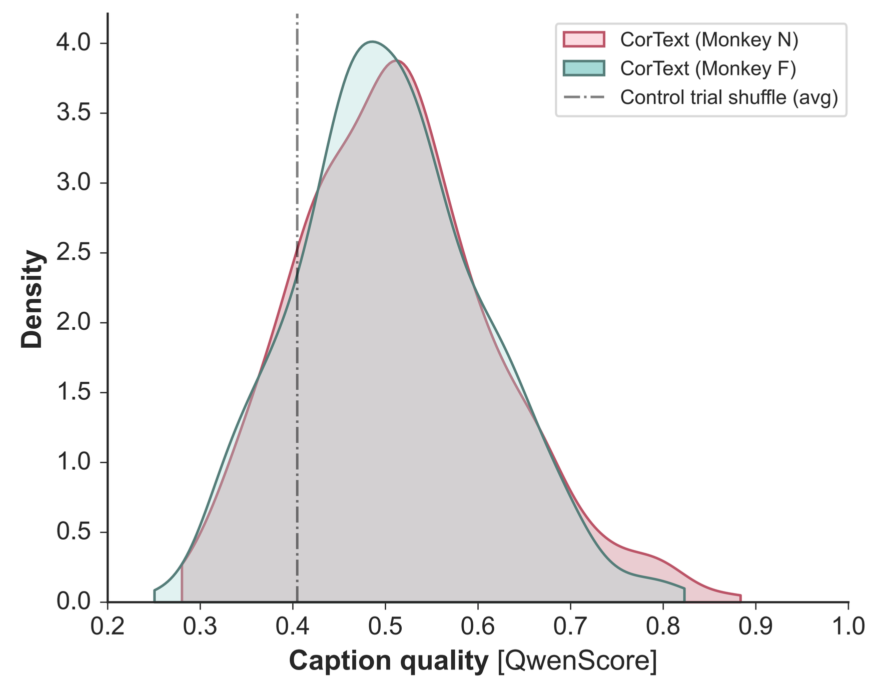


CorText caption: A collection of wooden dowel rods of varying lengths, neatly arranged on a wooden surface

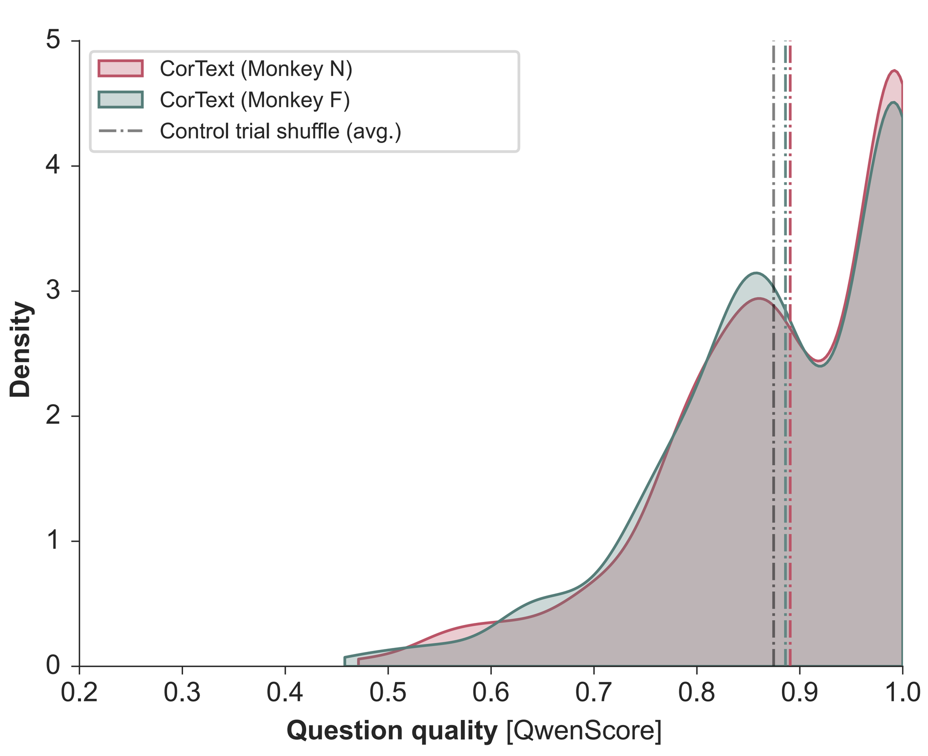
CorText caption: A small furry rodent possibly a chipmunk sitting on a rock looking directly at the camera.

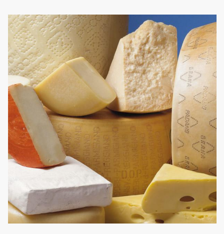
CorText caption: A pile of fresh whole almonds with their shells intact.





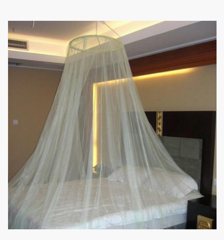
Detailed question-answering





CorText >> Control
Question: What are the main objects in the image made of?
CorText answer: The main objects are made of cheese.
Control answer: The main objects in the image are made of fabric.

CorText ~= control
Question: What material do the crayons appear to be made of?
CorText answer:The crayons appear to be made of wax.
Control answer: The crayons appear to be made of wax.



Control >> CorText
Question: How many pillows are visible on the bed?
CorText answer: There is one pillow visible on the bed.
Control answer: There are two pillows visible on the bed.

Challenges:
— What questions provide a good Q&A dataset for the THINGS stimuli, which primarily consist of simple, single-object scenes?
— How to best evaluate Q&A in this setting? It prevents the model from overfitting (vs training on only captions). Yet, it is challenging to evaluate, and control models can hallucinate: e.g, "what colour is the bamboo?" is easily answered without looking at the brain information. This is a challenge for future work regarding both dataset annotations, as well as evaluation metrics.

References

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